



Turbulence-supported Massive Star Envelopes

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Abstract

The outer envelopes of massive ($M \gtrsim 10 M_{\odot}$) stars exhibit large increases in opacities from forests of lines and ionization transitions (particularly from iron and helium) that trigger near-surface convection zones. One-dimensional (1D) models predict density inversions and supersonic motions that must be resolved with computationally intensive three-dimensional (3D) radiation hydrodynamic (RHD) modeling. Only in the last decade have computational tools advanced to the point where ab initio 3D models of these turbulent envelopes can be calculated, enabling us to present five 3D RHD ATHENA++ models (four previously published and one new $13 M_{\odot}$ model). When convective motions are subsonic, we find excellent agreement between 3D and 1D velocity magnitudes, stellar structure, and photospheric quantities. However, when convective velocities approach the sound speed, hydrostatic balance fails as the turbulent pressure can account for 80% of the force balance. As predicted by Henyey, we show that this additional pressure support leads to a modified temperature gradient, which reduces the superadiabaticity where convection is occurring. In addition, all five models display significant overshooting from the convection in the Fe convection zone. As a result, the turbulent velocities at the surface are indicative of those in the Fe zone. There are no confined convection zones as seen in 1D models. In particular, helium convection zones seen in 1D models are significantly modified. Stochastic low-frequency brightness variability is also present in the $13 M_{\odot}$ model with comparable amplitude and characteristic frequency to observed stars.

Unified Astronomy Thesaurus concepts: [Stellar physics \(1621\)](#); [Stellar convective zones \(301\)](#); [Stellar surfaces \(1632\)](#); [Stellar structures \(1631\)](#)

1. Introduction

The opacity (κ) increases associated with Fe and He excite near-surface convection zones (NSCZs) in massive star envelopes (Cantiello et al. 2009) across the Hertzsprung–Russell diagram (HRD). In many cases, one-dimensional (1D) modeling of these NSCZs predict transsonic convective velocities (v_c), and as the luminosity (L) approaches the Eddington value, $L_{\text{Edd}} = 4\pi GMc/\kappa$, produce unstable density inversions (Joss et al. 1973). These properties present a serious challenge to modeling the nature of convection in NSCZs with 1D approaches such as mixing length theory (MLT; Böhm-Vitense 1958), and motivate our extended suite of three-dimensional (3D) radiation hydrodynamic (RHD) simulations using ATHENA++ (Stone et al. 2020; Jiang 2021).

The 3D NSCZs show significant fluctuations in density (ρ) and radiative flux (F_r) that reduce the radiation pressure support as a result of porosity (Schultz et al. 2020). These density fluctuations also combine with transsonic velocities and propagate through the stellar photosphere to produce an optically thick wind in sufficiently high- L models (Jiang et al. 2015, 2018). As the plumes become optically thin near the photosphere, they impart significant Doppler broadening on the escaping photons, naturally generating the long-observed microturbulence measured in spectral line widths of massive stars (Cantiello et al. 2009; Schultz et al. 2023). The 3D surface convection also generates stellar brightness variability with

amplitudes and frequencies (Schultz et al. 2022) similar to observed massive stars (Bowman et al. 2020) in recent photometric surveys (e.g., TESS, Ricker et al. 2015) confirming 1D analyses (Cantiello et al. 2021).

In this paper, we utilize the variety in the optical depth at the Fe opacity peak (τ_{Fe}) in our ATHENA++ simulation suite to elucidate the diversity of 3D convection from regimes where energy is transported via convection to the more “lossy” realm where radiative energy transport dominates even though convection is vigorous. As discussed in Goldberg et al. (2021) and Jermyn et al. (2022), radiative transport becomes important when τ_{Fe} is less than the optical depth,

$$\tau_{\text{crit}} = \frac{P_r c}{(P_r + P_g) v_c}, \quad (1)$$

where P_r is the radiation pressure, P_g is the gas pressure, and c is the speed of light. When $\tau_{\text{Fe}} \ll \tau_{\text{crit}}$, convective plumes lose heat from radiative diffusion as they travel upwards causing a reduction in convective efficiency, and a smaller convective flux. For massive main-sequence and Hertzsprung Gap stars, $\tau_{\text{crit}} \gtrsim 1000$, allowing radiatively leaky NSCZs to exist inside an optically thick envelope. Such a realm of convection is implicitly included in the early works of Henyey et al. (1965), Ludwig et al. (1999), and Kuhfuss (1986), and those early estimates of the impact of the radiative losses guided our exploration.

Our focus here is two fold. First, we exhibit that for models where convective transport dominates, the transsonic velocities apply a significant pressure, $P_{\text{turb}} \propto \rho v_c^2$, that affects the outer envelope structure much like that seen in 3D models of red



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Table 1
3D Model Properties

Name	Masses		Temperature		Luminosities		Angular Size Ω_{sim} (sr/ π)	Optical Depth	
	M_{core} ($M_{\odot}$)	M_{env} ($M_{\odot}$)	$T_{\text{eff},F}^a$ (10^3 K)	T_{eff}^b (10^3 K)	L ($\log(L/L_{\odot})$)	$\Gamma_{\text{Edd,Fe}}^c$		τ_{Fe}	$\frac{\tau_{\text{Fe}}}{\tau_{\text{crit}}}$
M35ZAMS ^d	35	0.004	42	42	5.2	0.82	0.0013	515	0.02
M35MMS ^e	35	0.027	34	37	5.4	0.97	0.04	1084	0.25
M13TAMS	13	0.019	25	27	4.4	0.55	0.0096	3487	0.24
M80HG ^f	80	0.032	26	20	6.4	3.23	1.41	4300	2.3
M56HG ^g	56	0.036	15	9	6.2	3.38	1.41	25,686	8.1
Name	MESA Velocities		3D rms Velocities				Scale Heights		
	$v_{\text{MESA,Fe}}$ (km s^{-1})	$v_{\text{MESA,He}}$ (km s^{-1})	v_{Fe} (km s^{-1})	v_{He} (km s^{-1})	v_F (km s^{-1})	H_{Fe} ($R_{\odot}$)	H_F ($R_{\odot}$)	$H_{\text{tot,Fe}}$ ($R_{\odot}$)	$H_{\text{tot,F}}$ ($R_{\odot}$)
M35ZAMS	20	0	6.2	8.0	8.9	0.07	0.01	0.07	0.01
M35MMS	44	0	57	145	148	0.25	0.09	0.28	0.13
M13TAMS	11	0.01	8.3	12	19	0.11	0.02	0.11	0.02
M80HG	...	...	145	170	166	3.99	2.58	5.74	8.36
M56HG	...	...	81	132	161	13.34	4.83	16.45	31.12

Notes.^a Defined as the average temperature at the radius, r , that satisfies $L/4\pi r^2 = \sigma T_{\text{eff},F}^4$.^b Defined as the average temperature at the location where $\langle\tau\rangle = 1$.^c Eddington ratio at the Fe opacity peak.^d Model T42L5.0 in Schultz et al. (2022, 2023).^e Model T32L5.2 in Schultz et al. (2022) and T35L5.2 in Schultz et al. (2023).^f Model T19L6.4 in Jiang et al. (2018), Schultz et al. (2020, 2023).^g Model T9L6.2 in Jiang et al. (2018), Schultz et al. (2020).

supergiants (e.g., Goldberg et al. 2022). The temperature profiles in these 3D models can then be well explained (and predicted) by incorporating the impact of turbulent pressure using the Henyey et al. (1965) model. Similar work has shown P_{turb} plays a crucial role in the explosions of core-collapse supernovae (e.g., Couch & Ott 2015). Second, for hotter massive stars on the main sequence, the effects of turbulent pressure are much less (e.g., Grassitelli et al. 2015). However, these 3D RHD models exhibit convective motions far outside of the regions conventionally defined by 1D models and have detectable photometric variability.

2. Hydrostatic Balance in 3D Models of Turbulent Envelopes

Five 3D RHD ATHENA++ solar metallicity stellar envelope models were run to steady-state equilibrium, defined by reaching thermal equilibrium beneath the FeCZ. Table 1 details relevant quantities with the model names denoted by the core mass and phase of evolution: zero-age, middle of, and terminal age main sequence (ZAMS, MMS, and TAMS respectively), and Hertzsprung Gap (HG). The new model, M13TAMS, consists of a narrow wedge spanning >20 scale heights in both angular directions and a radial range that includes both the radiative region of the envelope below the convection zones (CZs) and the regions beyond the photosphere. Figure 1 shows the location in the HRD of these models relative to Modules for Experiments in Stellar Astrophysics (MESA; Paxton et al. 2011, 2013, 2015, 2018, 2019; Jermyn et al. 2023) tracks.

Schultz et al. (2020) showed that these 3D RHD models exhibit correlations that reduce the radiation pressure support, motivating our investigation of hydrostatic balance (HB). To quantify HB, the relative deviations, $(\rho g + dP/dr)/\rho g$, are plotted in Figure 2 using different choices for the pressure support. The x -axis is the pseudo-Mach number defined in Schultz et al. (2020) and increases outwards monotonically.

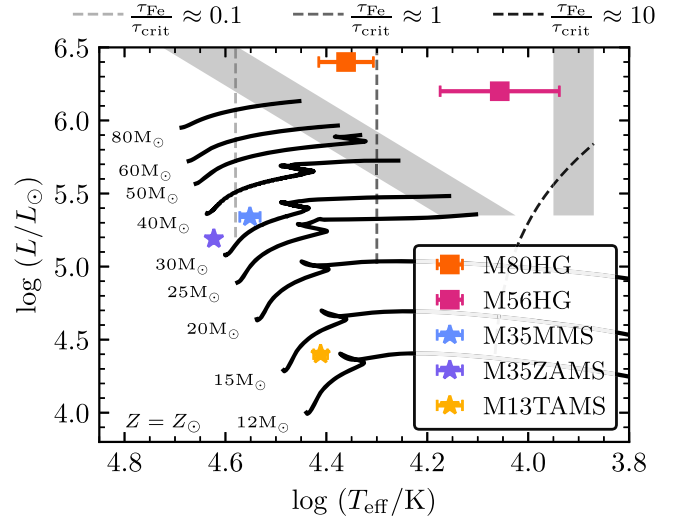


Figure 1. HRD showing the five 3D models (colored points). Squares represent models from Jiang et al. (2018) while the stars denote models from Schultz et al. (2022) and this work. The error bars show the difference between definitions of T_{eff} described in Section 3.1. The black lines are MESA models from Cantiello et al. (2021). The vertical dashed lines are approximate contours of $\tau_{\text{Fe}}/\tau_{\text{crit}}$ as calculated from the MESA models. Gray shaded regions show the S Dor and luminous blue variable (LBV) outburst instability strips from Smith et al. (2004).

Models M13TAMS and M35ZAMS are in HB with $P = \langle P_{\text{therm}} \rangle_V$, deviating by less than 5% throughout their envelopes. For the other three models, however, HB fails with M35MMS, M80HG, and M56HG reaching upwards of 41%, 73%, and 77% discrepancies at their surfaces respectively. These large deviations cannot be explained by the optically thick winds of M80HG and M56HG, as the advective velocity associated with the wind mass loss is less than the turbulent velocities inside the photosphere. The positive discrepancy

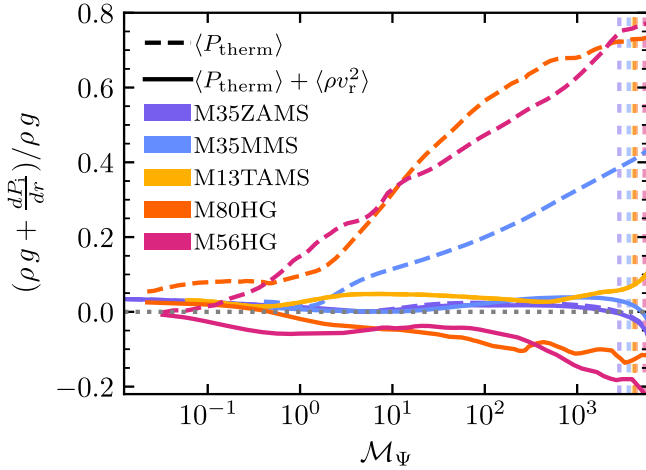


Figure 2. Fractional difference from HB for the average pressure of 3D models vs. $\mathcal{M}_\Psi$ including and excluding turbulent pressure (solid and dashed respectively). Vertical dashed lines on the right denote the location where the energy flux matches the photospheric criteria of σT^4 (e.g., $T = T_{\text{eff},F}$ from Table 1).

indicates additional pressure support is needed to reach force balance (FB) for these turbulent models.

As the rms velocities of the turbulence approach the sound speed, turbulent pressure given by $P_{\text{turb}} \equiv \langle \rho v_r^2 \rangle_V$ provides support. The choice of the radial velocity component, v_r , is motivated by expanding the radial component of the 3D spherical polar momentum equation where a ρv_r^2 term assists in balancing ρg (Goldberg et al. 2022). This additional pressure does not alter the comparison for M35ZAMS and M13TAMS as $P_{\text{turb}} \ll P_{\text{therm}}$ in these models. However the other three models show significant improvements in FB when P_{turb} is added. The deviations of M35MMS reduce to the same 5% as the two less turbulent models. The models with significant optically thick winds only experience $<20\%$ deviations within their photospheres, a factor of 4 improvement, and moreover, the deviations are in the expected direction. Accounting for P_{turb} in the luminous and massive models is required to correctly balance gravity in 1D stellar evolution models.

3. Comparing Averages of 3D Models to 1D Profiles

As 3D RHD models of massive star envelopes are computationally expensive, 1D models remain the most effective way to understand their evolution. However, 1D models require approximations that are not always verified with physically motivated 3D models. By comparing the 1D averages of the existing 3D models, we can verify the effectiveness of current 1D approximations. Spherical averages of the 3D models were calculated and compared to MESA models chosen to match their core masses and luminosities. Only the three wedge-like models (M35ZAMS, M35MMS, and M13TAMS) were able to be precisely matched as the global models (M80HG and M65H) have luminosities boosted by $\approx 50\%$, to yield faster convergence of their turbulent motions.

3.1. Extracting 1D Models from 3D Wedges

Distilling the turbulent 3D RHD models into 1D analogs is not as simple as using a volume weighted average. Using $\langle \rho \rangle_V$ and $\langle T \rangle_V$ to calculate the thermal pressure results in factor of 2 differences from $\langle P_{\text{therm}} \rangle_V$ near the surface, motivating a

reconsideration of conserved quantities when averaging 3D models to yield 1D profiles.

We chose the mass and internal energy contained within each radial shell to be the conserved quantities in the 3D to 1D translation. As the gas and radiation temperatures in each 3D cell deviate by $<2\%$, only one temperature is needed when calculating the internal energy therein,

$$E_{\text{int}} = \frac{3}{2}P_g + 3P_r = \frac{3k_B}{2\mu m_p}\rho T + a_r T^4, \quad (2)$$

where μ is the mean molecular weight. Thus we utilized the volume weighted averages of the internal energy, $\langle E_{\text{int}} \rangle_V$, and density, $\langle \rho \rangle_V$, to calculate a self-consistent average temperature, $\langle T \rangle_E$, using Equation (2). Replacing $\langle T \rangle_V$ with $\langle T \rangle_E$, reduces the fractional difference of the internal energy ($|E_{\text{int}}(\langle \rho \rangle_V, \langle T \rangle_V) - \langle E_{\text{int}} \rangle_V| / \langle E_{\text{int}} \rangle_V$) from upwards of 60% to numerical error. Using $\langle T \rangle_E$ to estimate the thermal pressure results in an average deviation of 10^{-3} with the maximal deviation staying $<5\%$ for all the models. This is a significant improvement compared to the average and maximal deviations of 5% and 60% respectively when using $\langle T \rangle_V$. Because of these improved agreements, $\langle T \rangle_E$ and $\langle \rho \rangle_V$ are taken to be the 1D averages of the 3D models.

Figure 3 compares $\langle T \rangle_V$ and $\langle T \rangle_E$ along with the typical spatial variation of T in a single temporal snapshot. The plotted regions of the models span from the innermost radius to reach thermal equilibrium to the outer radius where $\langle \tau \rangle_E = 1$ ($\langle \tau \rangle_E(r) \equiv \int_r^\infty \kappa(\langle T \rangle_E, \langle \rho \rangle_V) \langle \rho \rangle_V dr'$). In M13TAMS and M35ZAMS, $\langle T \rangle_E$ deviates from $\langle T \rangle_V$ by less than 1% while it changes by upwards of 23%, 20%, and 13% in M35MMS, M56HG, and M80HG respectively, signifying the turbulence generates correlations between T and ρ .

The overlapping or distinct locations of photosphere definitions, highlighted in Figure 3, arise from the varied topographies of these 3D RHD models. The 1D nature of M35ZAMS and M13TAMS leads to the location where $\langle \tau \rangle_E = 1$ coinciding within one radial cell of where $F = \sigma \langle T \rangle_E^4$. In contrast, the other three models display convective motions that reach or surpass the sound speed, resulting in strong turbulent motions at the photosphere and complex surface topography. The transsonic turbulence of M35MMS results in a 10% discrepancy in the predicted T_{eff} based on the choice of photospheric definitions, with the location where $\langle \tau \rangle_E = 1$ lying inside the flux defined surface. When turbulence becomes supersonic, as in M80HG and M56HG, the different T_{eff} definitions deviate by 30% and 40% respectively, with $F = \sigma \langle T \rangle_E^4$ occurring well below the location where $\langle \tau \rangle_E = 1$.

This contrast highlights the inherently 3D nature of these models. The turbulence causes complex surface topography with the radially integrated $\tau = 1$ surface spanning up to twice the stellar radius and encompassing both photospheric definitions plotted in Figure 3 (Schultz et al. 2023). The large topographic features yields factor of 2 temperature fluctuations, containing both definitions of T_{eff} in Table 1 and breaking the notion of a single T_{eff} or single radius photosphere. In order to show an approximation for the surface of the 1D averages of these 3D RHD models, $F = \sigma \langle T \rangle_E^4$ is chosen to define the 1D photosphere and future figures are truncated there.

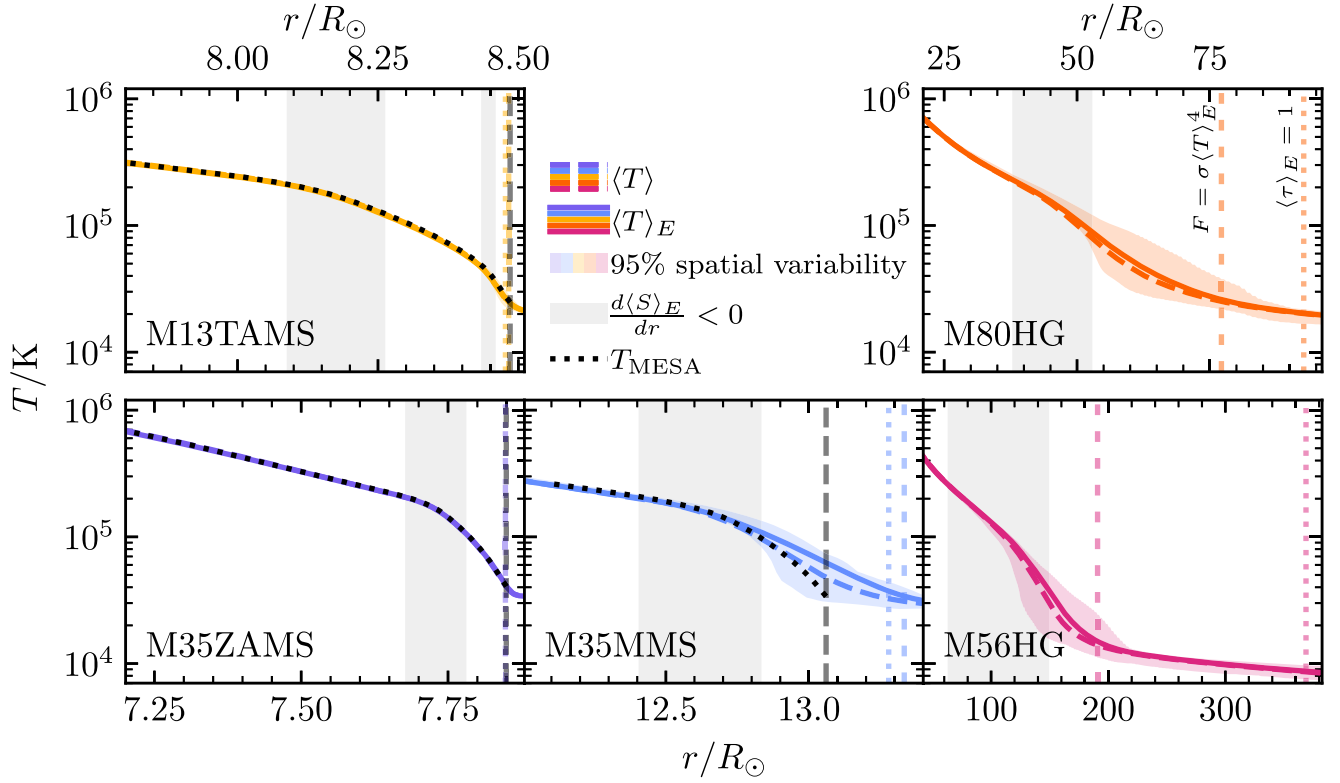


Figure 3. Comparison of radial profiles of the volume weighted average temperature, $\langle T \rangle_V$ (dotted lines), and the temperature extracted from the internal energy, $\langle T \rangle_E$ (solid lines), for the five models. The shaded color regions show the 95% spatial variability in a single temporal snapshot. Vertical colored lines show the radii where $\langle \tau \rangle_E = 1$ (dotted) and $L/4\pi r^2 = \sigma \langle T \rangle_E^4$ (dashed). The gray shaded regions denote where the average entropy gradient is negative, signifying a CZ. Temperature profiles from MESA models of similar stars to M13TAMS, M35ZAMS, and M35MMS are plotted in black dotted lines with their photospheric radii shown by the vertical gray dashed lines.

3.2. Comparing to MESA Models

Matching the solar metallicities of the 3D RHD models and adding exponential core overshooting with $f = 0.0014$, $f_0 = 0.004$ to smooth the HRD tracks, pre-main-sequence models for each core mass are modeled to ZAMS using the latest release of MESA (r22.11.1). These ZAMS models are then evolved utilizing the Henyey et al. (1965) MLT option, with the default parameters ($\alpha = 2$, $y = 1/3$, $\nu = 8$) until the $\langle T \rangle_E$, $\langle \rho \rangle_V$, and r of the base of the 1D analogs of the 3D RHD models are matched. All three models matched these conditions to within 5% as the base radii are well below the Fe opacity peak in a radiative region. The inlists used to create these models are available on Zenodo under an open-source Creative Commons Attribution license: doi:10.5281/zenodo.7972070.

The T (see Figure 3) and ρ MESA profiles agree with M35ZAMS and M13TAMS at all radii, including the photosphere. This is the first direct confirmation that 3D RHD ATHENA++ simulations with convective turbulence agree with 1D MESA models and highlights a region of the HRD where 1D stellar evolution models are sufficient to capture the stellar structure as well as the photospheric radii and temperatures. This region coincides with $\tau_{\text{Fe}} \ll \tau_{\text{crit}}$ or $L \ll L_{\text{Edd}}$ as in these limits the convection does not carry significant energy and cannot develop transonic turbulence. In contrast, the MESA analog to M35MMS is slightly more compact and hotter, with r_{photo} and $T_{\text{eff}} \approx 2\%$ different than the 3D estimates. Additionally, the 3D estimate exhibits a shallower temperature gradient through and above the FeCZ, causing the change in r_{photo} and T_{eff} . Despite the lack of MESA models to compare to, these discrepancies are likely to increase in the models that exhibit

stronger turbulent motions (i.e., models with more-massive cores, higher luminosities, and with $\tau_{\text{Fe}} \gtrsim \tau_{\text{crit}}$). It is clear that new prescriptions of turbulent heat transport are required to improve the 1D stellar evolutionary models in this regime.

The MESA models utilizing the default MLT parameters also predict convective velocities at the Fe opacity peak. Table 1 shows the difference between these velocities and the 3D rms velocities at both the Fe opacity peak ($v_{\text{MESA,Fe}}$ and v_{Fe} respectively) and the He opacity peak ($v_{\text{MESA,He}}$ and v_{He} respectively). The He opacity peak velocities disagree with those seen in the 3D models due to convective overshooting. At the Fe opacity peak, the velocities differ by 32% in M13TAMS, a factor of 3 in M35ZAMS, and 22% in M35MMS. Considering these comparisons utilize the same default MLT parameters for all the models, it is surprisingly accurate given the differences of the models.

Modifying the y and α MLT parameters for each MESA model can yield better agreement. Keeping $\alpha = 2$, the peak velocities in MESA's FeCZ agree when $y = 0.2$, $y = 0.1$, and $y = 1$ for M13TAMS, M35ZAMS, and M35MMS respectively. Alternatively, for $y = 1/3$ we find agreement with $\alpha = 1.7$, $\alpha = 1.3$, and $\alpha = 4.5$ for M13TAMS, M35ZAMS, and M35MMS respectively. These changes in y and α do not create significant changes in T_{eff} , r_{photo} , or the temperature profiles with all deviations being $< 0.5\%$ as radiation remains the dominant energy transport mechanism. Both y and α impact the estimates of the convective velocity; however, they impact different weak turbulence regimes. Reducing y is likely the correct modification when $\tau \ll \tau_{\text{crit}}$ (M35ZAMS) while

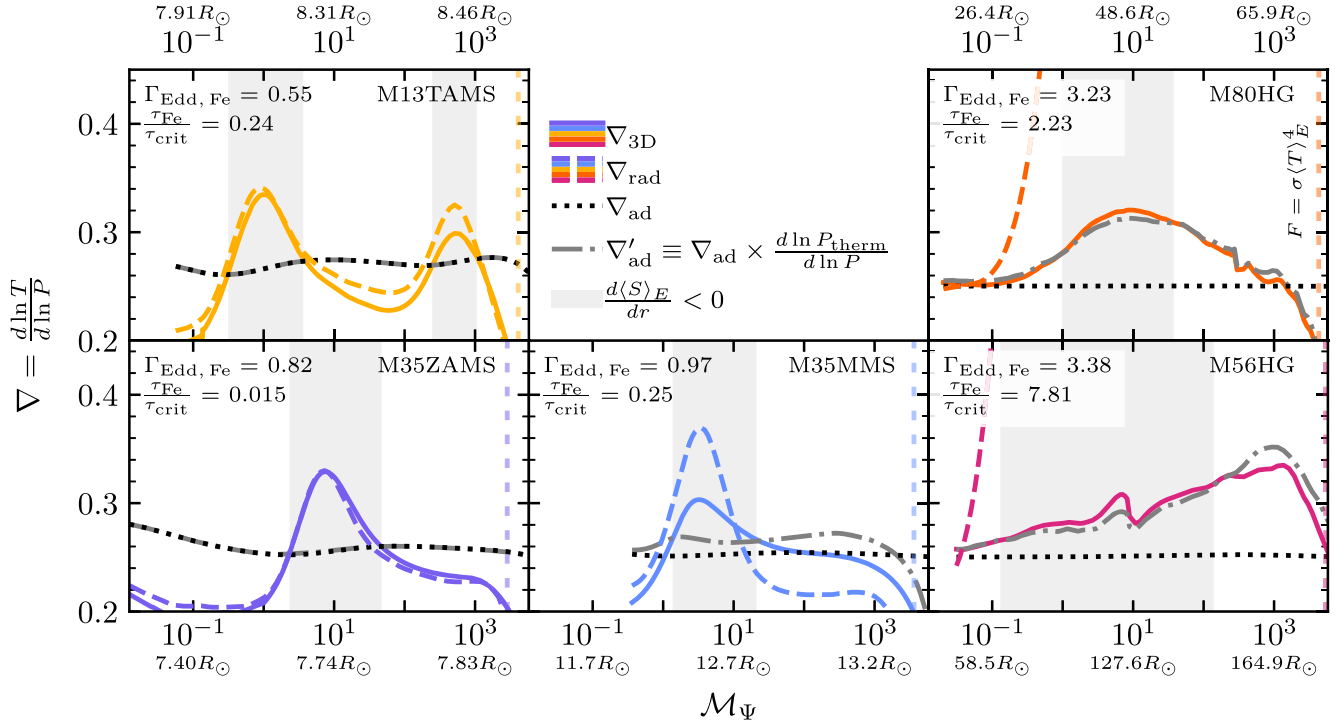


Figure 4. Comparison of temperature gradients for the five 3D RHD models including the realized temperature gradient, ∇_{3D} , radiative temperature gradient, ∇_{rad} , adiabatic gradient, ∇_{ad} , and the modified model adiabatic temperature gradient including P_{turb} , ∇'_{ad} (colored solid, colored dashed, black dotted, and gray dotted-dashed lines respectively). The vertical dashed lines are the same as in Figure 2. The gray shaded regions denote convectively unstable regions as in Figure 3.

reducing α is the better choice in the lower-luminosity regime where $L \ll L_{\text{Edd}}$ (M13TAMS).

3.3. Temperature Gradients in Turbulent Envelopes

Energy transport defines the temperature gradient, $\nabla \equiv \ln T / d \ln P$, in 1D stellar models, which is compared to two distinctly defined gradients. The first is ∇_{rad} , the temperature gradient required to carry the flux solely by radiative diffusion (see Equation (27) of Henyey et al. 1965). The second is the adiabatic temperature gradient, ∇_{ad} , which is often realized in efficient convection as plumes rise and fall nearly adiabatically. Figure 4 compares the 3D temperature gradient, ∇_{3D} , with ∇_{rad} and ∇_{ad} . For M13TAMS and M35ZAMS, ∇ is well approximated by ∇_{rad} further agreeing with the $\tau \ll \tau_{\text{crit}}$ picture. The convective plumes lose enough heat from radiation as they rise that the convective flux is negligible and thus $\nabla \approx \nabla_{\text{rad}}$.

In more turbulent models, a different picture unfolds with ∇_{3D} deviating from both ∇_{rad} and ∇_{ad} . Below the Fe opacity peak in M35MMS, ∇ follows ∇_{rad} to within 5%; however, there is a departure from ∇_{rad} in and above the CZ that is also seen in M80HG and M56HG. Models M80HG and M56HG have $L \approx L_{\text{Edd}}$ and exhibit $P_r \gg P_{\text{gas}}$ resulting in $\nabla_{\text{rad}} \gg \nabla_{\text{ad}}$. As both are convectively efficient ($\tau \gtrsim \tau_{\text{crit}}$) the convective flux grows until the velocities approach the sound speed, limiting the energy transport. The resulting ∇_{3D} lies between ∇_{rad} and ∇_{ad} suggesting a different mechanism of convective inefficiency, aside from radiative losses.

However, as discussed earlier, these transsonic velocities imply a significant P_{turb} , suggesting the need for a modified ∇_{ad} . In this regime, Henyey et al. (1965) used the chain rule to

account for the role of turbulent pressure in ∇_{ad} by defining

$$\nabla'_{\text{ad}} = \nabla_{\text{ad}} \times \frac{d \ln P_{\text{therm}}}{d \ln P}, \quad (3)$$

where $P = P_{\text{therm}} + P_{\text{turb}}$. For M13TAMS and M35ZAMS, ∇'_{ad} is equivalent to ∇_{ad} but for the others there is a noticeable change. In fact, when ∇'_{ad} is used, the superadiaticity ($\nabla_{3D} - \nabla'_{\text{ad}}$) of M80HG and M56HG is reduced significantly implying the convection is truly efficient and follows the modified model adiabatic temperature gradient that includes the impact of P_{turb} . Additionally, the definition of the CZs, where $dS/dr < 0$, agree identically with the regions where $\nabla > \nabla'_{\text{ad}}$ rather than the conventional ∇_{ad} in all the models. Thus when models have significant turbulent pressure, 1D models must utilize ∇'_{ad} to properly estimate the CZs' properties.

4. Conclusions

We present five 3D RHD ATHENA++ models that highlight the impact of turbulence in massive star envelopes. When the stellar luminosities are sufficiently high and NSCZs are adequately optically thick as to develop transsonic convective velocities, the ram pressure of the turbulence can account for nearly 80% of the pressure support. This additional pressure support modifies the temperature change of an adiabatically rising plume, reducing the estimated superadiaticity and confirming that convection is efficient. Estimating the turbulent pressure for 1D models using physically motivated prescriptions will be vital for improving estimates of stellar observables. That being said, it is important to remember that turbulence makes photospheres inherently 3D and care should be taken when interpreting 1D photospheric values in this realm.

The presence of ubiquitous stochastic low-frequency (SLF) brightness variability from massive stars has caused significant debate about their origin. The SLF variability was shown to naturally originate from the NSCZs in the more-massive simulations presented in this work (M35ZAMS, M35MMS, M80HG, M56HG; Schultz et al. 2022). However, the formation mechanism for lower-mass and less-luminous stars remains a mystery, with core internal gravity waves (e.g., Bowman et al. 2020) and stellar wind variability (e.g., Krtićka & Feldmeier 2018) also proposed as possible sources. Our introduction of the lower-mass model M13TAMS allows us to address what occurs in lower-mass stars. Following Cantiello et al. (2021), the characteristic frequency, ν_{char} , of M13TAMS is 1.5 day^{-1} , consistent with observations of similar stars (Bowman et al. 2020). The rms amplitude of the $\log L$ fluctuations from M13TAMS (accounting for the $\sqrt{n}$ geometric factor of combining n wedge models across the stellar surface) is $\approx 130 \mu\text{mag}$, comparable to the amplitude of variability in observed stars. Thus NSCZs in all five of our massive star envelope models exhibit SLF variability on timescales consistent with measured ν_{char} and at levels comparable to those observed. The finite runtimes of 3D models (due to computational limitations) inhibit our predictions for frequencies considerably less than ν_{char} , with a different minimum frequency for each model (e.g., in Schultz et al. 2022, no prediction was made for $\nu < 0.5 \text{ day}^{-1}$).

These 3D RHD models are computationally expensive (taking 3000 Skylake cores 4 days to run 1 model day), limiting our ability to populate the HRD and generate a P_{turb} prescription to be used in 1D stellar evolution. However, our work has highlighted two clear regimes. In the weak turbulence regime, in the lower left of the HRD, the NSCZs are either too radiatively lossy (M35ZAMS) or have sufficiently low luminosities (M13TAMS) to impact the stellar structure. The opposite is true for the efficient luminous regime in the upper right of the HRD (M80HG and M56HG) where turbulence dominates the envelope. M35MMS is the only model currently in the transition between these two regimes. For this reason we have only attempted to compare the existing models to their MESA counterparts and note the impacts of the turbulent NSCZs. We are excited to see that 1D stellar evolution models agree with 3D RHD simulations in the weak turbulence regime and as more models are computed, the transition between spherically symmetric and inherently 3D envelopes may be mapped and 1D turbulent pressure prescriptions developed to improve stellar evolution models.

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